

Scalability & Future Plan

Date	15 March 2024
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No database in current version	No permanent student history or reports.	High
2	AI processing needs compute resources	Long audio may take more time.	High
3	No login/dashboard in current version	Limited multi-user classroom use.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local/low-volume use	Deploy on VPS/Render and cache AI models.
2	Data Storage	Local folders only	Add SQLite/PostgreSQL and cloud file storage.
3	Performance	Model runs during analysis	Use smaller Whisper model and background tasks.
4	Security	No authentication	Add login, role access, and secure file handling.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Add database and login system	1-2 months	Stores student history and reports.
Phase 3	Teacher dashboard and analytics	2-3 months	Helps educators track progress.
Phase 4	Cloud deployment and scaling	3-4 months	Improves access and performance.